ENGINEERING PROJECT / 6 NOVEMBER 2026

AeroTrace

Causal diagnosis for selected cross-system failures in an A320 simulation

3	6	190
native mechanism classes	sealed version 2 episodes	software tests passed

The engineering question

When generator output and hydraulic pressure fall in the same incident, which causal configurations fit the signals available to the debugger? AeroTrace shows the evidence for each candidate and preserves competing histories when their predicted effects agree.

Qantas Flight 32 motivates this distinction. The crew reported a falling Green hydraulic fluid level indication. The Australian Transport Safety Bureau documented monitoring wiring damage and found no Green fluid loss in the later examination. This project tests the distinction with selected native A320 simulator interventions. [1]

How it works

Stage	Responsibility
Capture	Collect timestamped public observations from FlightGear. Save commands and readiness checks separately.
Evidence	Assess generator output, bus voltage, Blue quantity indication and Blue pressure. Preserve missingness and onset brackets.
Diagnosis	Compare required and conditional effects, record contradictions, and display current impact separately from incident history.
Review	Use the same diagnostic rules in live and recorded views. Apply sealed evaluation targets after public-only prediction.

Version 1 is the default two-mechanism catalogue. Explicit version 2 adds a pressure-supply path and the corresponding paired explanations. Both use a finite compatibility model.

Source loss with bus voltage retained

A fresh FlightGear A320 session at about 10,171 feet and 250 knots recorded 53 complete polls around a paired generator 2 and Blue electric pump intervention.

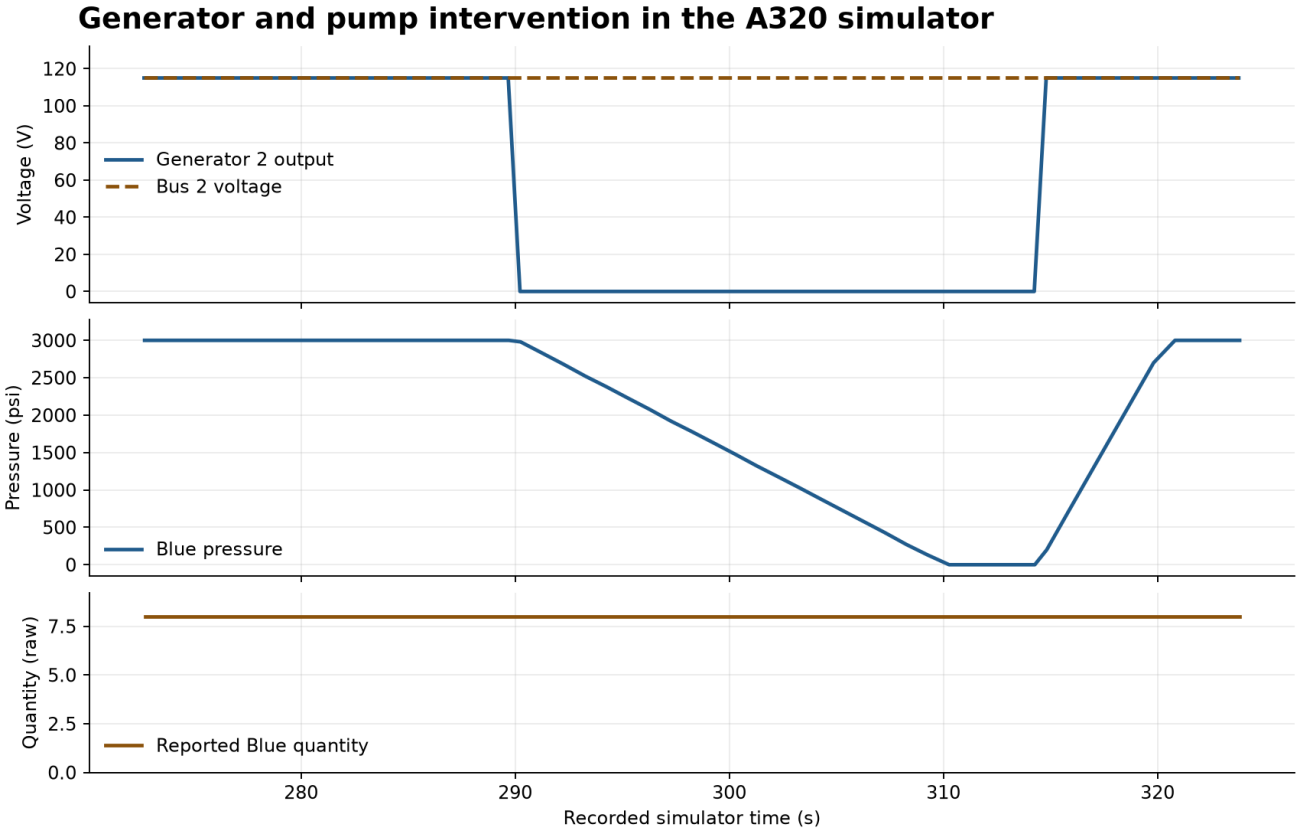


Figure 1. Public observations plotted at their per-channel simulator timestamps. Voltage is shown in volts (V), pressure in pounds per square inch (psi), and quantity in raw model units.

Observation	Recorded effect
Generator 2 output	Reached zero volts while both monitored alternating current buses stayed near 115 volts.
Blue pressure and quantity	Pressure reached zero from 3,000 psi. The quantity indication stayed at 8 raw units.
Recovery	Pressure and voltage recovered. All three supported flags were verified clear.

At samples 19 through 21 the generator source-path configuration fit the available evidence. By sample 22 the hydraulic evidence supported two pressure-pair histories. The live and offline final diagnoses agreed, and replay kept the incident after recovery.

A sealed test across two flights

The version 2 protocol fixed the catalogue, temporal method, detector, inference and scorer hashes, known development inputs and planned assignments before collection. Each prediction ran on the public file before its private scoring target was written.

Flight	Observed baseline	Reference	Pump only	Pair
A	8,134 ft / 240 kt	32 polls	53 polls	53 polls
B	6,096 ft / 231 kt	32 polls	53 polls	53 polls

Baseline altitude is shown in feet (ft) and airspeed in knots (kt).

All six final candidate sets matched their sealed targets. Both pairs retained the independent and grouped configurations. The 276 polls are correlated within the two flight groups. The two groups share the aircraft model, airport and weather setup.

What timing contributed

The controlled comparison used identical observed facts for alert-only, dependency and temporal dependency outputs. Timing produced zero discrimination changes in the native version 1 and version 2 pilots. The version 1 pilot covered four declared cases in one flight.

Pump-only replay input	Dependency	Temporal dependency
Original native observations	Pressure supply	Pressure supply
Early false quantity decline	Inventory loss	Inventory loss
Late false quantity decline	Inventory loss	Poor model fit

The software stress changes the quantity indication alone. An early false reading fits both methods. A late false decline contradicts the inventory-first timing rule. The display describes the quantity indication and leaves physical fluid loss unresolved.

The NASA diagnosis evaluation framework informed the separation of observation, private scenario information and scoring. This project uses its own finite candidate-set targets. [2]

Inspect the result from source

Extract the review bundle and run the following command from its source folder with Python 3.11 or later. Enter the command as a single line.

```
python -B -m aircraft_debugger.sim_panel --replay
results/pressure-pair-20261106-01/pair/observations.jsonl --model v2 --port 8769
```

Open http://127.0.0.1:8769. Move from sample 17 through sample 22, then to the final sample. Expand both compatible pressure-pair candidates and compare the restored readings with the incident evidence.

Software and update cost

A fresh standard-library environment passed 190 tests and reproduced the paired report, timeline and model byte for byte. The original sealed version 1 sources also reproduced the earlier paired prediction and score.

In a 1,200-poll synthetic extension of the active pair, the measured 95th-percentile local adapter update was 490 milliseconds before airborne context reuse and 80 milliseconds after. Host background loads differed. The timing covers local processing and snapshot copy.

Evidence boundaries

The simulator properties support bounded compatibility and measured system impact. The physical initiating history, component identity, independent sensor confirmation and broader aircraft capability remain unresolved. The live engine explicitly stops inference at 1,200 polls. The recorded demo uses the included public file and runs with Python alone.

References and included evidence

[1] Australian Transport Safety Bureau, AO-2010-089 final report, Appendix B, printed page 218. Final report.

[2] Kurtoglu et al., Towards a Framework for Evaluating and Comparing Diagnosis Algorithms, sections 3.3 and 3.4. NASA DASHlink copy.

The source bundle includes README.md, docs/ENGINEERING-REPORT.md, the paired and prospective observations, the frozen protocols, separate scoring records and full software checks.